Enabling the industry of tomorrow

stargate hydrogen

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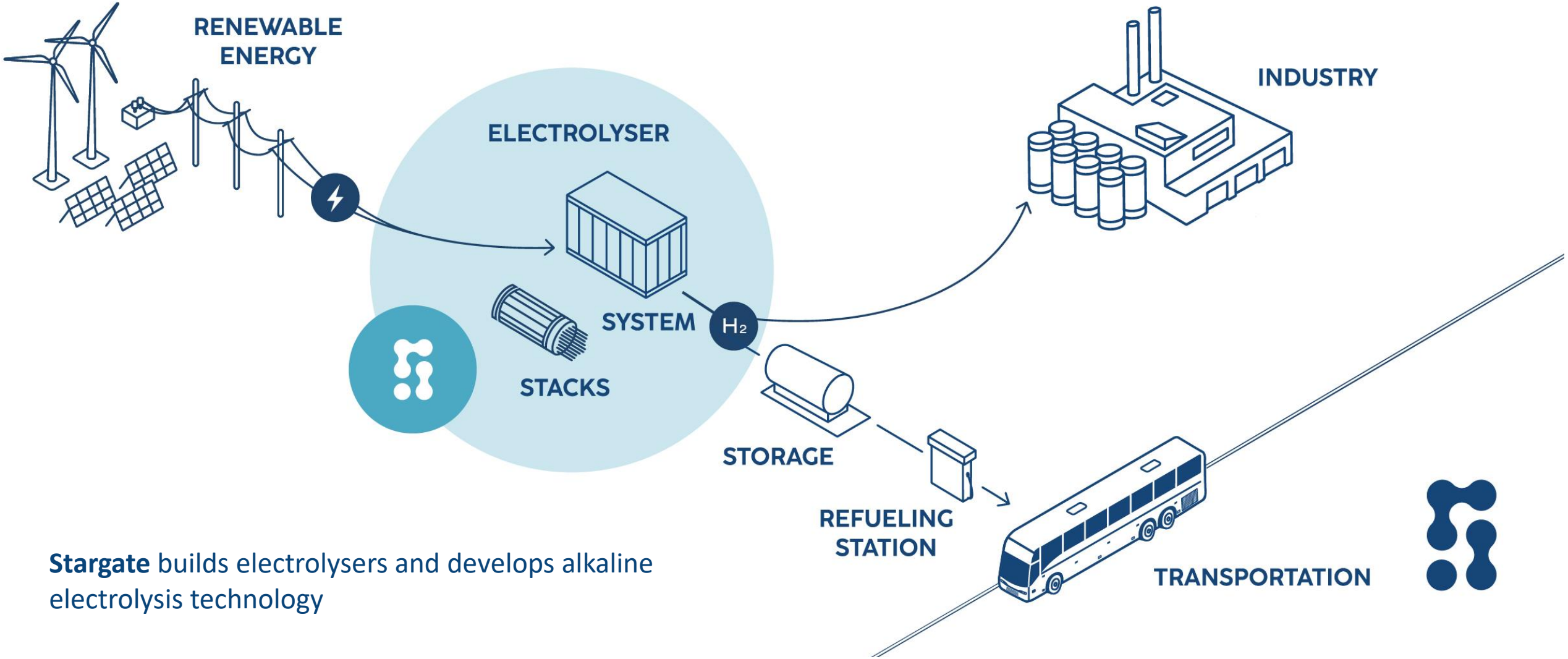
About us

Rethink decarbonisation by hydrogen



Stargate builds electrolysers and develops alkaline electrolysis technology

Stargate Hydrogen in numbers

We are manufacturer of pressurised Alkaline electrolyser stacks and turn-key systems based in Tallinn Estonia.

Our uniqueness lies in the development of catalyst material Stardust that enables us to provide the most efficient products on the market.

- Employees: 52+
- Product groups: 2
- Patents pending: 7
- Annual production volume: 50 MW
- Ongoing electrolyser projects: 4
- EU strategic technology funding: 60 MEUR
- Production capacity from 2027: 1 GW

Stargate is backed by investors

- One of the leading family offices in the region, focused on energy and sustainability-related investments.
- Core holding of UGI is Utilitas Group which is the largest renewables developer in the region with 7GW of offshore wind in development.
- A strong balance sheet exceeding €400 million, allowing to support Stargate over the long run.

- Global Cleantech 100 company.
- Technology leader in fast energy storage for automotive, transportation, grid, and industrial applications.
- Founded in Estonia operating in Germany.

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Products

Products of Stargate Hydrogen

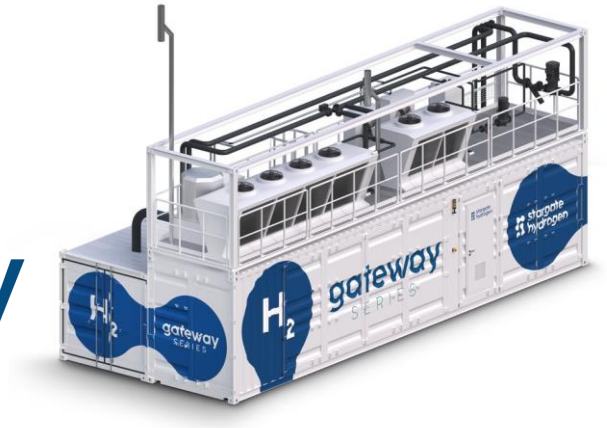
stellar
SERIES

Pressurized Alkaline electrolyser stacks for third party system integration

Specifications:

- Modular design
- From 20 to 100 Nm^3/h
- From 0,05 to 0,5 MW

gateway
SERIES



Turn-key pressurized Alkaline Electrolyser systems for hydrogen production

Specifications:

- Modular design
- From 200 to 2000 Nm^3/h
- From 0.5 – 10 MW

Gateway Electrolysers

Gateway electrolysers are containerised turn-key green hydrogen production systems using pressurised alkaline technology.

Stacks and Balance of Stack

- Electrolyser stacks
- Gas separation system
- Power rectifier
- Control system
- Lye loop

Balance of plant

- Nitrogen purge system
- Safety system
- Back-up power system
- Vent stacks
- System (rectifier, stack, gas) cooling

Optional extras

- Gas purification system
- Water purification system
- Process air compressor
- Waste-heat reuse system
- Low temperature package
- MV Transformer & switchgear
- Inter-container process and electrical connections
- Hydrogen storage system
- Compressor system

gateway
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Leading performance

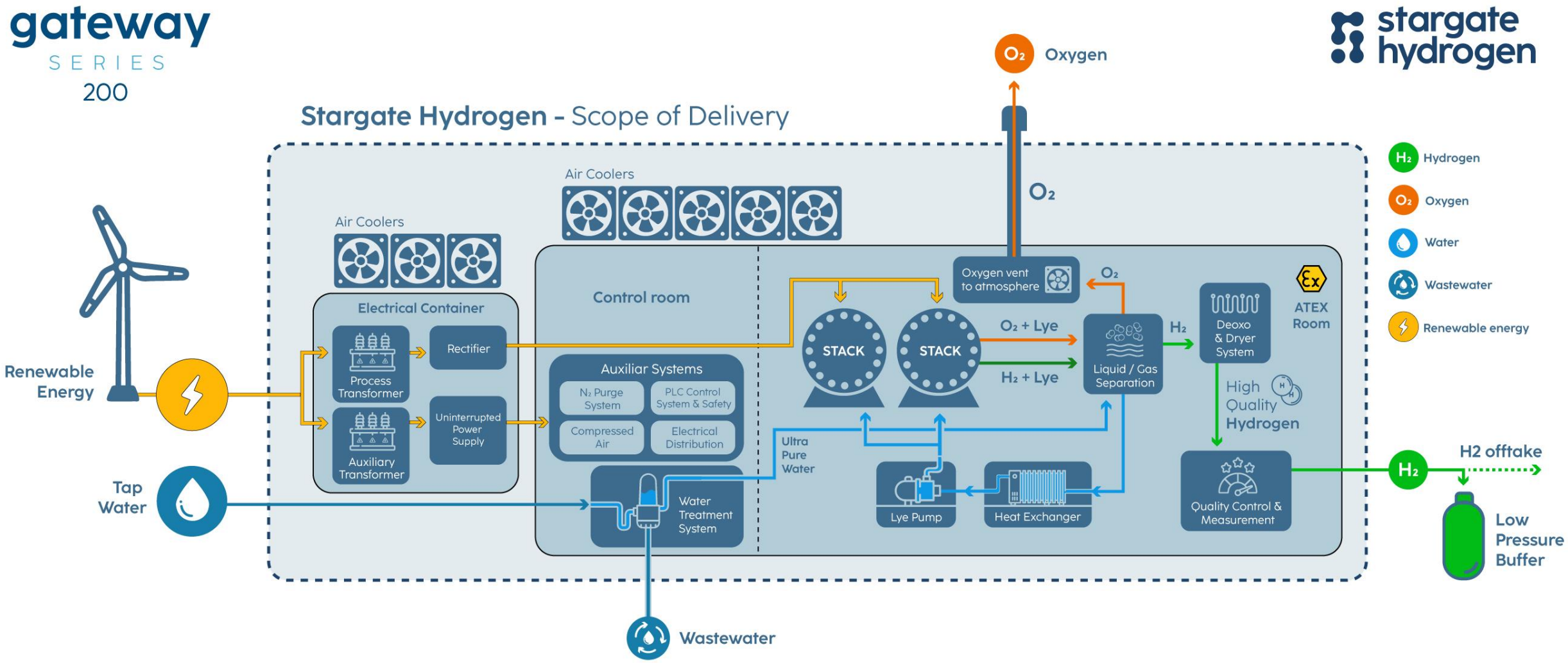
Full Integration support

Low CAPEX

High Purity Hydrogen

High Efficiency

Gateway 200 - Diagram

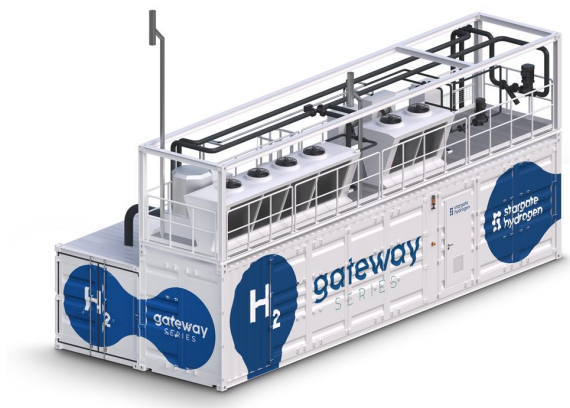


Technical specifications

Product	Gateway 200	Gateway 400	Gateway 800	Gateway 1600	Gateway 2000
Hydrogen hourly production rate [Nm³/h]	200	400	800	1600	2000
Hydrogen daily production rate [kg/day]	432	864	1728	3456	4320
Hydrogen pressure [barg]	30	30	30	30	30
Hydrogen purity [%] *	> 99.999%	> 99.999%	> 99.999%	> 99.999%	> 99.999%
Oxygen purity [%]	> 99.7%	> 99.7%	> 99.7%	> 99.7%	> 99.7%
Installed electrical power [MVA]	1.2	2.4	4.8	9.6	12
Stack consumption [kWh/Nm³]	4.28	4.28	4.28	4.28	4.28
System efficiency (HHV) [%]	74.5%	74.5%	74.5%	74.5%	74.5%
System efficiency (LHV) [%]	62.9%	62.9%	62.9%	62.9%	62.9%
Operating range [%]	20-100%	10-100%	5-100%	5-100%	5-100%
Electrolyte	KOH	KOH	KOH	KOH	KOH
Electrical interface	0.4 kV, 50/60 Hz, > 0.9 PF	0.4 kV, 50/60 Hz, > 0.9 PF	0.4 kV, 50/60 Hz, > 0.9 PF	0.4 kV, 50/60 Hz, > 0.9 PF	0.4 kV, 50/60 Hz, > 0.9 PF
Pure water requirement [l/h]	164	328	656	1312	1640
System installation location	Outdoors (containerized)	Outdoors (containerized)	Outdoors (containerized)	Outdoors (containerized)	Outdoors (containerized)
Ambient temerature range [°C] **	-20 to +40	-20 to +40	-20 to +40	-20 to +40	-20 to +40
Communication interface	Ethernet/Modbus	Ethernet/Modbus	Ethernet/Modbus	Ethernet/Modbus	Ethernet/Modbus

* Target purity achievable with optional purification system

** Target temperature range available with optional extra package - Standard: +5 to +40 °C



gateway
SERIES

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Alkaline electrolyser stacks for system integrators

Our pressurised alkaline electrolyser stacks are available with a 6-month lead time and come with a 10-year performance guarantee.

Specifications:

- From 20 to 100 Nm^3/h
- Performance guarantee for 10 years
- Engineering support

Customizeable configurations

Performance guarantee for 10 years

Integration support

Fast delivery 6 months lead time

Up to 100 Nm^3/h

High Efficiency

Stack configuration options

Standard Stack:

- Rated power: 0,5 MW
- Hydrogen production rate: 100 Nm^3/h

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Operational packages

Operation conditions of the stack can be adjusted focusing on CAPEX or efficiency

Max Efficiency: stack operating conditions are set to provide lowest cost of hydrogen. Suitable for projects with high price of electricity.

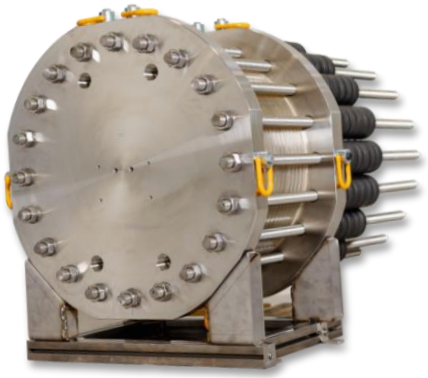
Low CAPEX : stack operating conditions are set to provide lowest CAPEX. Suitable for projects with low price of electricity.

Balanced: stack operating conditions are balanced between price of hydrogen and CAPEX.

KW size modularity

By adjusting cells in the stack, we can provide different kw sizes of the stacks:

- Electricity consumption: from 50 to 500 kw
- Hydrogen production rate: from 10 to 100 Nm^3/h



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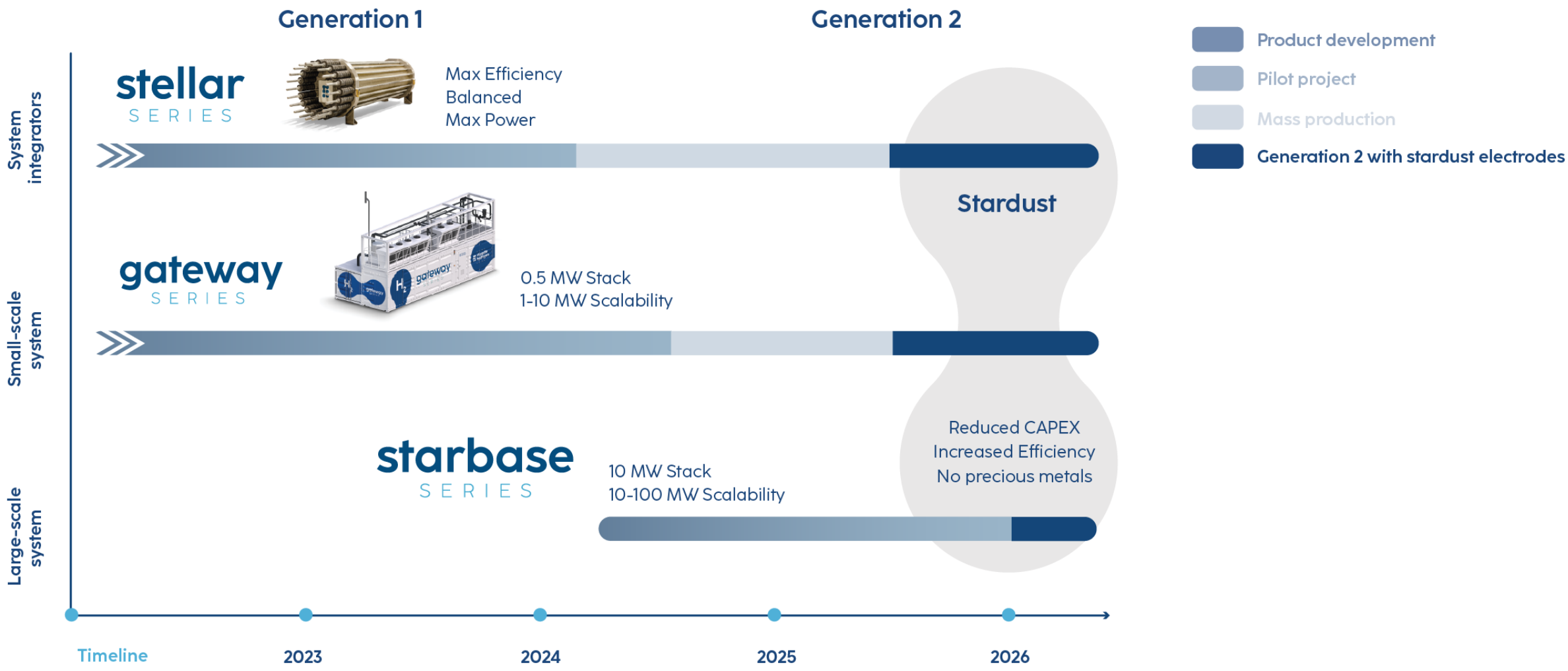
SERIES

Technical specifications

Product	Max Efficiency	Balanced	High Power
Hydrogen hourly production rate [Nm³/h]	100	100	100
Hydrogen daily production rate [kg/day]	215	215	215
Hydrogen pressure [barg]	32	32	32
Hydrogen purity [%]	>98%	>98%	>98%
Oxygen purity [%]	>98%	>98%	>98%
Stack Consumption [kWh/Nm³]	4.59	4.74	4.84
Stack Consumption [kWh/kg]	51.07	52.74	53.85
Stack Operating temperature [°C]	80-90	80-90	80-90
Stack rated voltage - BOL [V]	227	189	160
Stack rated voltage - EOL [V]	264	213.4	178.2
Stack rated current [A]	2027	2507	3022
Stack minimum current [A]*	1150	1150	1150
Stack rated power - BOL [kW]	460	474	484
Stack rated power - EOL [kW]	535	535	539
Stack minimum operating point [%]*	55%	45%	38%
Stack efficiency (HHV) [%]	77.2%	74.7%	73.2%
Stack efficiency (LHV) [%]	65.2%	63.1%	61.8%

* Lower minimum load point on request.

Product roadmap



Go to Market Strategy - System

gateway SERIES

Product

Systems

Target customer

Utility / RE Developer

- Interested to become hydrogen producer & seller or producer of hydrogen derivatives.
- Limited experience of installing electrolysers before / small in-house team.

Industrial customer

- Replacing grey H_2 with green.

USP

- Pressurized high-efficiency system.
- Made in EU.
- Flexible scope to meet customer needs.
- Lowest levelized cost of hydrogen in the long-term.

Examples

UTILITAS

fortum

Customer roadmap

Pilot projects
1 – 2 MW

Industrial use-cases
2 – 10 MW

Large-scale projects
100+ MW

Long-term target:
Preferred EU supplier for 100MW+ projects of target customer

2023 | 2024 | 2025 | 2026 | 2027

Go to Market Strategy - Stack

stellar SERIES

Product

Alkaline Stacks

Target customer

System Integrators

- Strong in-house process engineering experience.
- Interested to enter green hydrogen value chain or to replace existing stack supplier.
- Limited in-house experience with electrolysers.

USP

- Comprehensive integration support.
- Configurable stack design.
- Pressurized high-efficiency stack technology.
- Lower risk of leakages due to novel sealing solution.
- Lowest levelized cost of hydrogen in the long-term.

Examples

ROCKFIN®

MILANI SPA
IMPIANTI ELETTRICI E TECNOLOGICI

Customer roadmap

Proof of concepts
0.1 – 1 MW

Recurring revenue
0.5 – 5 MW projects

Long-term target:
100 MW of stack sales per integrator per year

2023 | 2024 | 2025 | 2026 | 2027

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RnD of Alkaline Technology

Research & Development

Research at Stargate is mainly concentrated to the three main technologies inside of an Alkaline electrolyser system.

Cell technology – material innovation

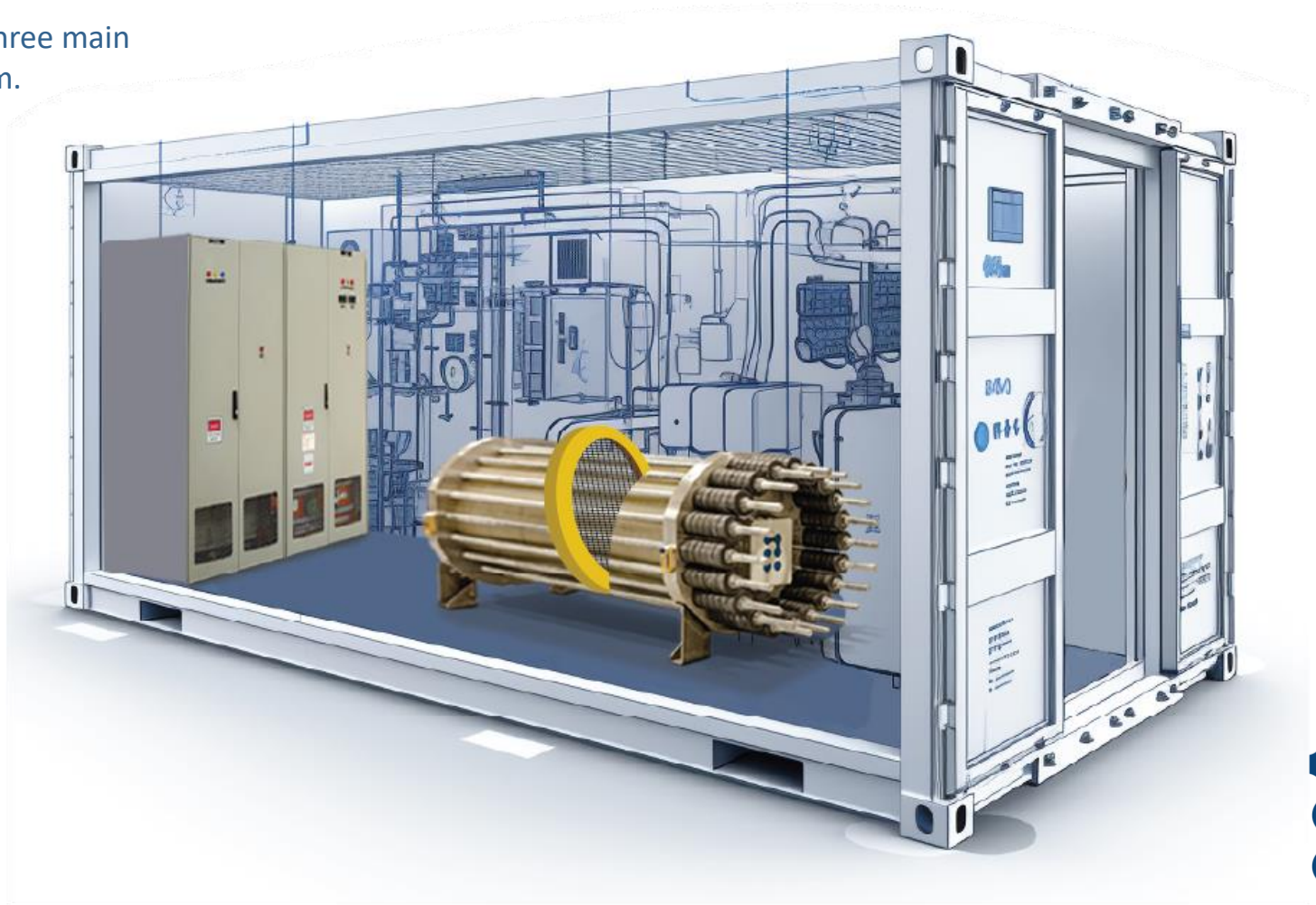
Stargate is developing innovative catalyst material Stardust that is working on high current densities and is free from PGM materials. Goal is 45 kwh/kg hydrogen.

Stack technology – stack optimization

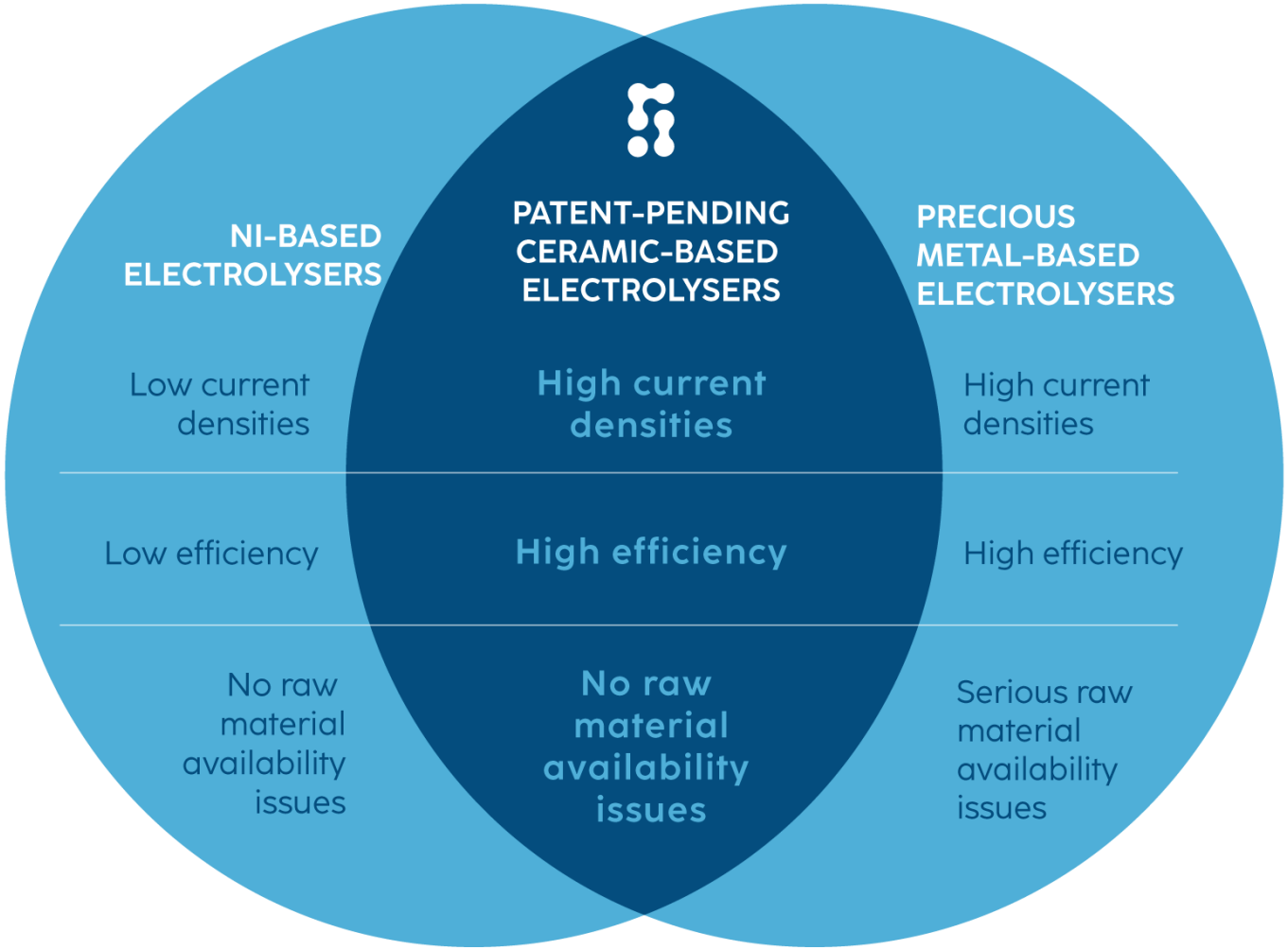
Stargate is optimising the work parameters to increase efficiencies and lifetime of the electrolysis stacks.

Power control – control system innovation

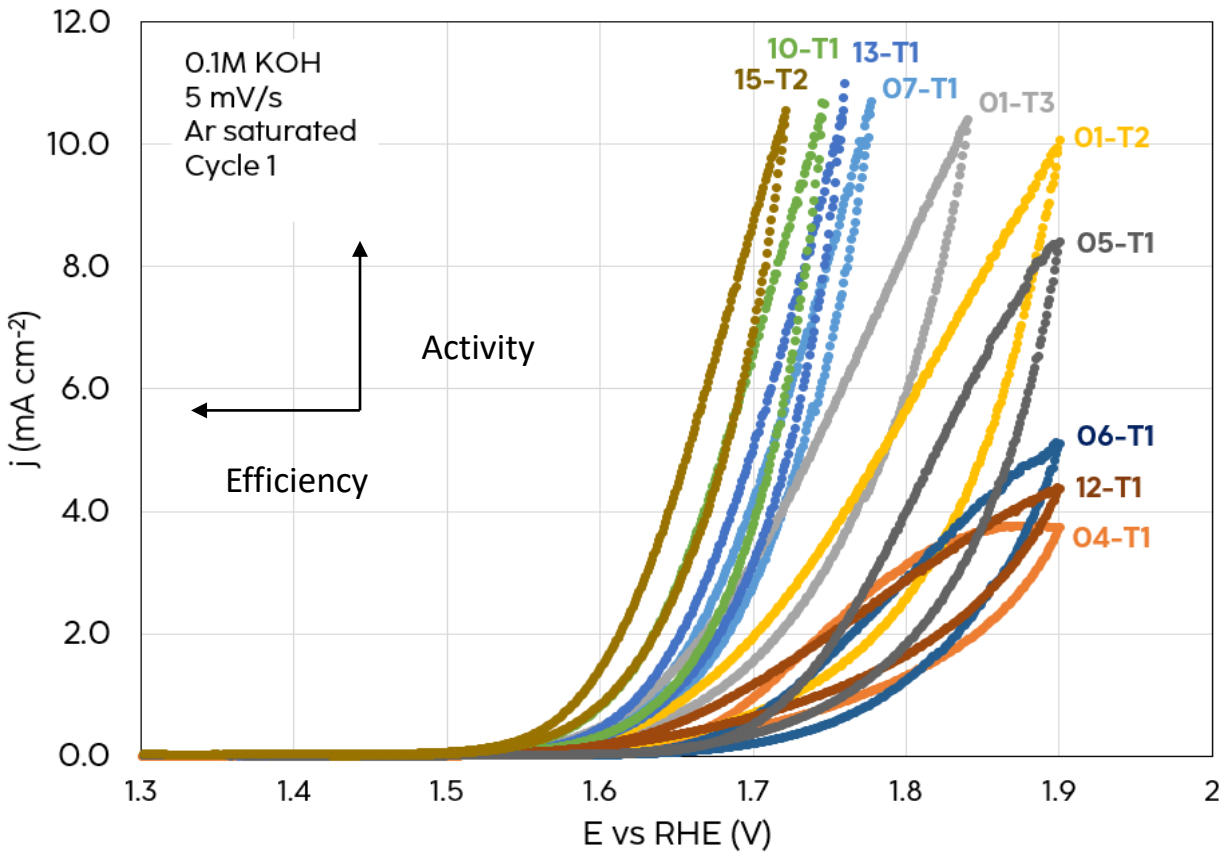
Stargate is developing new products for higher power levels challenging power control systems and configurations to work at gigawatt-scale.



Beyond the state-of-the-Art Technology

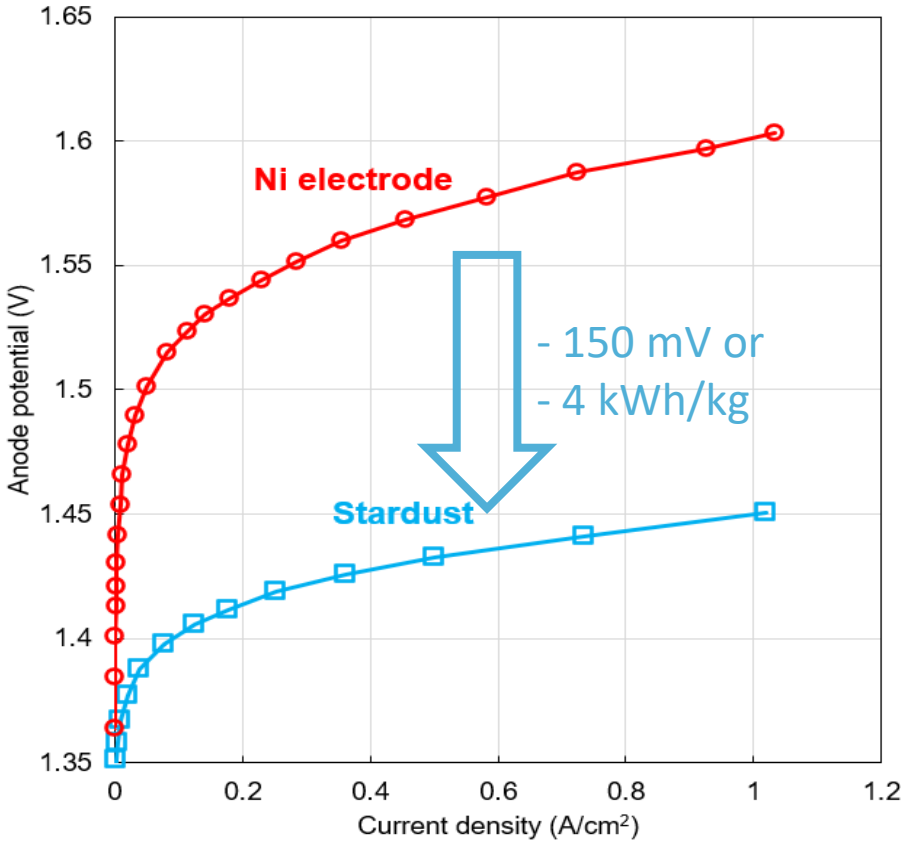


Focus on Catalyst Material development



More than 100 materials evaluated

Electrode performance validation by Fraunhofer Institute



"The results of the 3-electrode measurement of the coated electrode are impressive."

Dr. rer. nat. Christian Immanuel Bernäcker
Group Manager Electrochemical Technology

Fraunhofer IFAM

Test conditions: 30 wt% KOH, 80°C, 1 atm, two-compartment cell, three-electrode setup, Stardust or regular Ni electrode used as anode, Pt sheet used as counter and reference electrodes (via Luggin capillary).

R&D Partners

TAL TECH

University of Tartu

University of St Andrews

TU/e Eindhoven University of Technology

UCLouvain

AGFA

Fraunhofer IFAM

ZSW

PLOCAN Plataforma Oceánica de Canarias

ABB

NACO

enel Green Power

PERMASCAND

Ha Fundación para el Desarrollo de las Nuevas Tecnologías del Hidrógeno en Aragón

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References

Commercial progress

Product	Client	Country	Timeline	LOI
10 kW Stack	Rockfin	Poland	Delivered July 2023	100 MW
10 kW Stack (mockup)	Non-disclosed customer	Austria	Delivered September 2023	N/A
100 kW Stack	Milani	Italy	Delivered November 2023	50-100 MW
30 kW Stack	Non-disclosed customer	France	Delivered December 2023	N/A
Engineering support services	Next heat	Germany	Delivery in Q1 2024	100 MW
10x 5kW Stack	ABB	Finland	Delivery in Q2 2024	N/A
2x 0.5 MW Stack	Non-disclosed customer	Germany	Delivery in August 2024	N/A

Commercial progress

Product	Client	Country	Timeline	LOI
1 MW Gateway System	Utilitas	Estonia	Delivery in Q3 2024	N/A
1 MW Gateway System	PLOCAN	Spain	Delivery in Q2 2025	N/A
0.5 MW Stack	Biga H_2	Turkey	Delivery in Oct-2024	50 MW

Hydrogen for mobility in Estonia

- 1 MW electrolyser
- 2 refuelling stations
- Waste heat re-use in bio-mass CHP plant
- Operational by Q3 2024

Goal

Owner of the project is Baltic´s largest renewable developer. Goal is to gain experience in pilot project before scaling to larger volumes.

Hydrogen use case

- Hydrogen will be used by local taxi fleet

Stargate Hydrogen´s part

- Delivery of electrolyser, compressor, tube trailer and dispensers

Green H_2 for heating in Spain

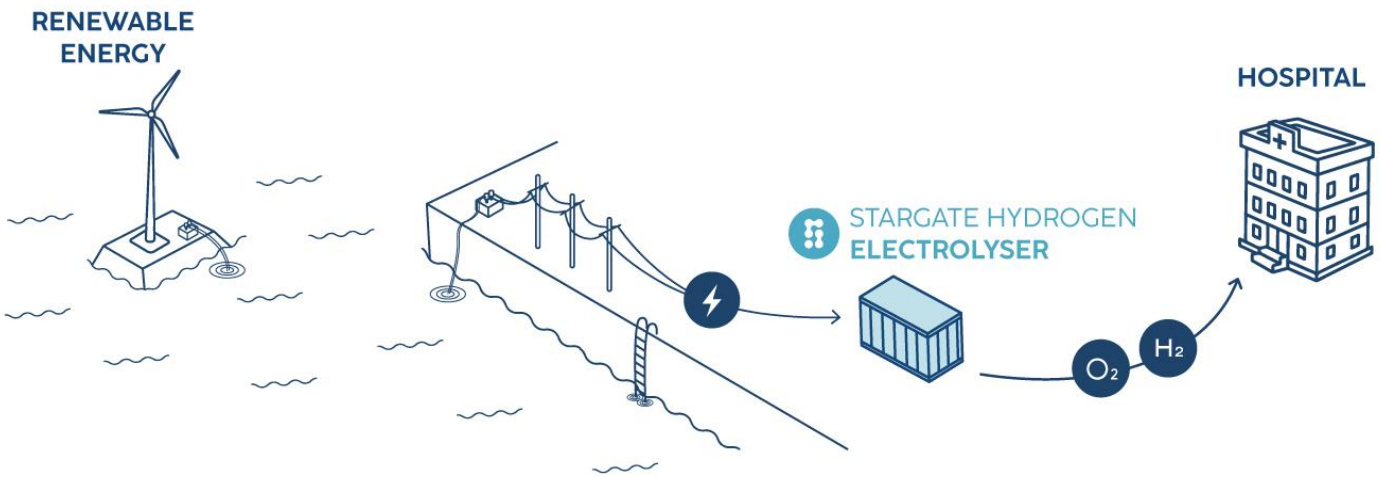
Goal: The owner of the project is the local municipality. The goal is to test hydrogen production from wind in the off-shore conditions with the option to scale in the future.

Stargate Hydrogen´s part

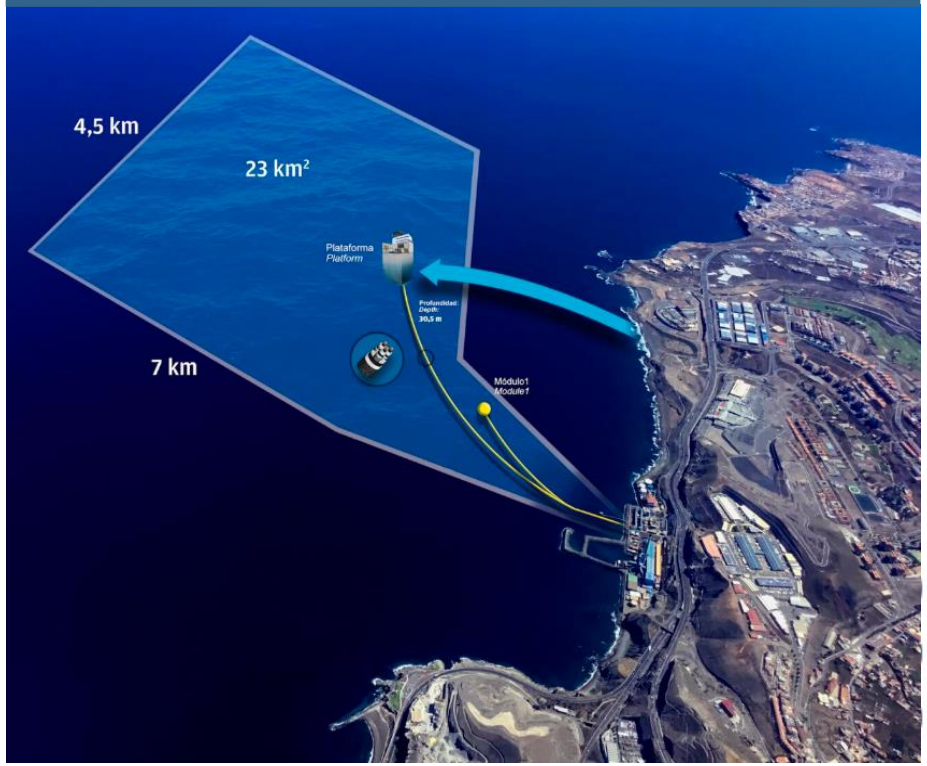
- Delivery and commissioning of the electrolyser

Hydrogen use case

- Hydrogen and oxygen will be used at local hospital



- 1 MW electrolyser
- Working on off-shore conditions
- Power from offshore wind turbine
- Signed in September 2023



Stack integration in Poland

Customer case
Rockfin integrates Stargate Hydrogen´s stacks into an electrolyser systems to provide turn-key hydrogen production solutions for its industrial customers.

About Rockfin
Rockfin is Europe´s leading engineering company specializing in the design, manufacture, testing and servicing of oil and gas systems.

Value to Rockfin
By utilizing Stargates technology, Rockfin enters into new business field of hydrogen production. Rockfin increases its expertise and value offering by providing its customers new portfolio of hydrogen production solutions.

Stargate Hydrogen´s part

- Delivery of electrolysis stacks
- Technical integration support

Stack integration in Italy

Customer case
MILANI integrates Stargate Hydrogen´s stacks into an electrolyser systems to provide turn-key hydrogen production solutions for its industrial customers.

About Milani
MILANI is an Italian engineering company specializing in the engineering, fabrication and installation of Skid mounted units for process industry and electrical plants.

Value to MILANI
By utilizing Stargates technology, MILANI enters into new business field of hydrogen production. MILANI increases its expertise and value offering by providing its customers new portfolio of hydrogen production solutions.

Stargate Hydrogen´s part

- Delivery of 100 kw electrolysis stack for pilot integration project
- Technical integration support

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Let's keep in touch

Scan the QR code to visit our website.

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